Hackathon Problem Statement – 2 (MECON)
AI-Driven Risk-Aware Production Scheduling & Predictive Maintenance Optimization Platform

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1. Background and Context

Industrial Common Facility Centers (CFCs) and heavy manufacturing clusters operating steel, iron, and fabrication equipment depend on high-value machines functioning under dynamic load and operating conditions.

However, several structural challenges persist:

- Unplanned machine failures cause significant downtime and production loss.
- Maintenance scheduling is often interval-based rather than condition-based.
- Production scheduling does not incorporate machine health risk.
- Throughput maximization, downtime minimization, and maintenance cost control must be balanced simultaneously.
- Operational decisions lack predictive modelling and structured scenario evaluation.

There is a need for a software-based, AI-driven internal decision-support system that integrates machine learning, simulation modelling, and multi-objective optimization to enable intelligent production planning under uncertainty.

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2. Main Objective

The objective of this challenge is to design and develop a software-based, AI-driven production intelligence platform that:

- Predicts machine failure probability or remaining useful life using supervised learning.
- Generates machine health scores based on operational indicators.
- Simulates maintenance timing impact on downtime, cost, and production loss.
- Performs risk-aware job scheduling under operational constraints.
- Supports dynamic trade-offs between throughput, downtime risk, and maintenance cost.
- Ensures transparency by clearly presenting allocation, deferral, and maintenance decisions.

The solution must be entirely software-based and must use synthetic datasets generated by participating teams following the mandatory schema.

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3. System Architecture Requirements

The solution must include two integrated layers:

Layer 1: Operator Intelligence Interface

This module must:

- Accept machine and job inputs based on the mandatory dataset schema.
- Implement a supervised machine learning model to predict Failure_Label or Remaining_Useful_Life.
- Generate a Machine Health Score (0–100 scale).
- Identify high-risk machines.
- Simulate at least two maintenance timing scenarios (e.g., immediate vs delayed).
- Estimate projected downtime, production loss, and cost under each scenario.
- Allocate jobs to machines without scheduling conflicts.
- Incorporate machine risk into job allocation logic.
- Provide explainability for failure predictions and maintenance recommendations.

Layer 2: Management Optimization Dashboard

This module must:

- Aggregate all machine and job data.
- Accept adjustable operational weight sliders for:
  - Throughput Priority Weight
  - Downtime Risk Minimization Weight
  - Maintenance Cost Minimization Weight
- Define and implement a weighted objective function.
- Optimize job allocation such that:

    - Machine capacity constraints are respected.
    - Maintenance windows are enforced.
    - Job deadlines are satisfied.
    - Each job is assigned to only one machine.
- Display:
    - Optimized production schedule
    - Predicted total throughput
    - Predicted downtime hours
    - Predicted maintenance cost
    - High-risk machines
    - Deferred jobs with justification
    - Comparative outputs when weights are adjusted

The system must demonstrate dynamic scenario simulation when priority weights are modified.

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4. Mandatory Dataset Structure (Schema Only)

Participants must generate synthetic datasets following the structure below.

Machine Dataset (Minimum 10 Machines)

Identification Fields

- Machine_ID
- Machine_Type
- Installation_Year
- Machine_Age

Operational Indicators

- Total_Run_Hours
- Avg_Load_Percentage
- Daily_Operating_Hours
- Capacity_Per_Hour

Condition Indicators

- Avg_Temperature
- Avg_Vibration
- Energy_Consumption_Rate
- Last_Maintenance_Days

Historical Indicators

- Failure_History_Count
- Downtime_Hours_Last_Year
- Maintenance_Cost_Last_Year

Target Variable (Mandatory for ML)

- Failure_Label (0/1)
  or
- Remaining_Useful_Life_Hours

Job Dataset (Minimum 50 Jobs)

- Job_ID
- Required_Machine_Type
- Processing_Time_Hours
- Load_Requirement_Percentage
- Priority_Level (1–5)
- Deadline_Hours
- Revenue_Per_Job

Maintenance and Cost Parameters

- Preventive_Maintenance_Cost
- Corrective_Maintenance_Cost
- Downtime_Cost_Per_Hour
- Replacement_Cost

- Maintenance_Duration_Hours

All assumptions used in synthetic data generation must be clearly documented.

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5. Phase-Wise Evaluation and Checkpoints

Total Marks: 100

Phase 1: Data Design and Assumptions (15 Marks)

Checkpoint Requirements

- Correct implementation of mandatory schema.
- Logical synthetic data generation.
- Realistic operational and failure distributions.
- Clear definition of prediction target.
- Clear documentation of assumptions.

Evaluation Criteria

- Data realism: 5 marks
- Structural adherence: 5 marks
- Assumption clarity: 5 marks

Phase 2: Failure Prediction Model (20 Marks)

Checkpoint Requirements

- Implementation of a supervised machine learning model.
- Prediction of Failure_Label or Remaining_Useful_Life.
- Generation of a Machine Health Score.
- Model evaluation metrics (Accuracy, F1-score, RMSE or relevant measures).
- Feature importance or explainability mechanism.

Evaluation Criteria

- Model design and feature engineering: 10 marks
- Evaluation and validation approach: 5 marks

- Explainability and integration into system logic: 5 marks

Phase 3: Maintenance Impact Simulation (20 Marks)

Checkpoint Requirements

- Implementation of at least two maintenance timing scenarios.
- Downtime estimation logic linked to failure risk.
- Production loss estimation logic.
- Cost modelling for preventive and corrective maintenance.
- Clear scenario comparison outputs.

Evaluation Criteria

- Simulation correctness: 10 marks
- Integration with ML predictions: 5 marks
- Projection clarity and consistency: 5 marks

Phase 4: Multi-Objective Scheduling Optimization (25 Marks)

Checkpoint Requirements

- Implementation of conflict-free job allocation logic.
- Clear objective function definition incorporating weighted priorities.
- Integration of adjustable operational weight sliders.
- Respect for machine capacity, deadlines, and maintenance constraints.
- Logical ranking and selection of machine–job allocations.

Evaluation Criteria

- Optimization correctness: 10 marks
- Weight integration and responsiveness: 10 marks
- Decision reasoning and transparency: 5 marks

Phase 5: Dashboard, Transparency and System Integration (20 Marks)

Checkpoint Requirements

- Functional operator interface.
- Functional management dashboard.
- Real-time weight-based scenario simulation.
- Display of optimized production schedule.
- Display of high-risk machines.
- Display of deferred jobs or machines with justification.
- Clear visualization of throughput, downtime, and cost metrics.

Evaluation Criteria

- Interface usability and clarity: 5 marks
- Transparency and justification: 5 marks
- Scenario simulation functionality: 5 marks
- End-to-end system integration: 5 marks

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6. Constraints

- Software-only solution.
- Synthetic data only.
- Single planning horizon (e.g., one-week simulation).
- Each job must be assigned to only one machine.
- Jobs cannot be split.
- Each machine may undergo at most one maintenance cycle within the planning horizon.
- Operational constraints must be strictly enforced.

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7. Expected Outcome

The final solution should demonstrate:

- Supervised machine learning-based failure prediction.
- Data-driven maintenance timing simulation.
- Risk-aware production scheduling.

- Multi-objective operational optimization.
- Throughput–downtime–cost trade-off analysis.
- Transparent and explainable decision-support for industrial operations.